

A PROJECT REPORT ON
A COMPARATIVE STATISTICAL ANALYSIS OF LIFE EXPECTANCY BETWEEN
DEVELOPED AND DEVELOPING COUNTRIES USING R

Submitted to
DEPARTMENT OF STATISTICS

In partial fulfillment of the requirements
for the award of M.sc., degree in statistics

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CERTIFICATE

Certified that this is entitled “**A COMPARATIVE STATISTICAL ANALYSIS OF LIFE EXPECTANCY BETWEEN DEVELOPED AND DEVELOPING COUNTRIES**” is a bonafied record of project work done by **_ROJA.V, ADDEPALLI ANUSHA**, M.Sc., Statistics, Department of Statistics, S.V. University, Tirupati for partial fulfillment of the requirement for the reward of M.Sc. degree.

Place: Tirupati

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Examiner-2

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ROJA.V
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DECLARATION

We hereby declare that this dissertation titled “**A COMPARATIVE STATISTICAL ANALYSIS OF LIFE EXPECTANCY BETWEEN DEVELOPED AND DEVELOPING COUNTRIES**” is a bonafied and genuine project work carried out by us under guidance of **Prof. B. SAROJAMMA**, Department of Statistics, Tirupati. This work is purely a new contribution of us and it was not used anywhere earlier for the purpose of any degree or award.

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ABSTRACT

Life expectancy is one of the most important indicators used to measure the overall well-being and development of a country, reflecting health conditions as well as economic and social progress. This study first examines the trend in life expectancy from 2016 to 2024 using linear regression analysis to evaluate changes over time. The results indicate a statistically significant upward trend in both developed and developing countries, although a considerable gap between the two groups persists throughout the study period.

This project presents a comparative statistical analysis of life expectancy between developed and developing countries using data from 2016 to 2024. The dataset includes life expectancy along with key economic and health-related variables such as GDP, healthcare expenditure, and mortality indicators. Countries were classified into two groups based on development status to examine structural differences.

To determine whether a significant difference exists between the two groups, One-Way ANOVA was applied. Two-Way ANOVA classification was further used to analyze the combined effect of country status and year on life expectancy. Correlation analysis was conducted to explore the relationships between life expectancy and selected socio-economic variables. In addition, multiple regression model were developed to predict life expectancy and assess the impact of multiple influencing factors, with model accuracy evaluated using appropriate statistical measures.

The findings reveal that developed countries consistently maintain higher life expectancy compared to developing countries. Economic growth, healthcare improvements, and reductions in mortality rates were found to play a significant role in enhancing longevity. Overall, the study highlights the persistent disparity between nations and underscores the importance of strengthening economic and health policies to reduce global inequalities.

Keywords: Life Expectancy; Developed Countries; Developing Countries; Trend Analysis; One-Way ANOVA; Two-Way ANOVA; Correlation Analysis; Multiple Regression; Socio-Economic Factors; Health Indicators; Predictive Modeling.

INTRODUCTION

1. Meaning and Concept of Life Expectancy

Life expectancy is one of the most important indicators used to understand the overall health and development of human societies. It refers to the average number of years a person is expected to live, assuming that current patterns of mortality remain the same throughout their lifetime. In simple words, life expectancy shows how long people in a particular country or region usually live.

Life expectancy is not just a biological measure; it is also a reflection of social, economic, environmental, and technological progress. A higher life expectancy usually indicates better healthcare systems, improved living standards, higher literacy rates, good nutrition, and a clean environment. On the other hand, a low life expectancy often reflects poverty, poor healthcare, malnutrition, lack of education, and environmental problems.

From ancient times to the present day, life expectancy has changed dramatically. Human beings once struggled to survive beyond a few decades, but today many people live well into their seventies, eighties, and even nineties. This remarkable change shows how human civilization has progressed in medicine, science, technology, and social organization.

2. Life Expectancy in Ancient Times

In ancient and prehistoric periods, life expectancy was very low compared to modern times. Early humans lived in harsh conditions where survival itself was a challenge. They faced many dangers such as wild animals, natural disasters, lack of food, and deadly diseases. There were no hospitals, medicines, or scientific knowledge to treat illnesses.

In prehistoric times, the average life expectancy was around 20 to 30 years. Many people died at a young age due to hunger, infections, injuries, and childbirth complications. Infant mortality was extremely high, meaning many children did not survive beyond the first few years of life.

In ancient civilizations such as Egypt, Greece, Rome, and India, life expectancy slightly improved but remained low, usually between 20 and 35 years. Although these civilizations

developed agriculture, architecture, and early medical knowledge, they still lacked advanced healthcare and sanitation. Epidemics, wars, and poor living conditions frequently reduced the population.

During the Middle Ages, life expectancy slowly increased to around 30–40 years, but it was still far from modern standards. Plagues like the Black Death, frequent wars, and poor hygiene caused millions of deaths. People had limited understanding of germs and diseases, so prevention and treatment were almost impossible.

Thus, in ancient times, life expectancy was mainly affected by:

- High infant mortality
- Infectious diseases
- Lack of medical knowledge
- Poor nutrition
- Unsafe living conditions
- Wars and conflicts

3. Life Expectancy in Modern Times

The modern era has witnessed a revolutionary increase in life expectancy. With the advancement of science and technology, humans have developed effective ways to fight diseases and improve living conditions. The discovery of vaccines, antibiotics, and modern medical techniques has saved millions of lives.

In the early 20th century, the global average life expectancy was about 30–35 years. However, by the 21st century, it has increased to more than 70 years in many parts of the world. In some developed countries, people live beyond 80 years on average.

Several factors contributed to this improvement:

- Development of modern medicine and surgery
- Availability of vaccines and antibiotics
- Clean drinking water and sanitation systems
- Better nutrition and food production
- Improved education and awareness

- Technological innovations in healthcare
- Strong public health policies

Modern societies also focus on preventive healthcare, which helps people avoid diseases before they become serious. As a result, mortality rates have decreased, and life expectancy has increased significantly.

However, despite global progress, life expectancy is not equal everywhere. There are large differences between developed and developing countries, which highlights global inequality in health and living standards.



4. Comparison of Life Expectancy: Developed and Developing Countries

Life expectancy varies greatly between developed and developing countries. Developed countries such as Japan, Canada, Germany, the United Kingdom, and the United States have high life expectancy due to advanced healthcare systems and high living standards. In contrast, many developing countries in Africa and parts of Asia and Latin America have lower life expectancy.

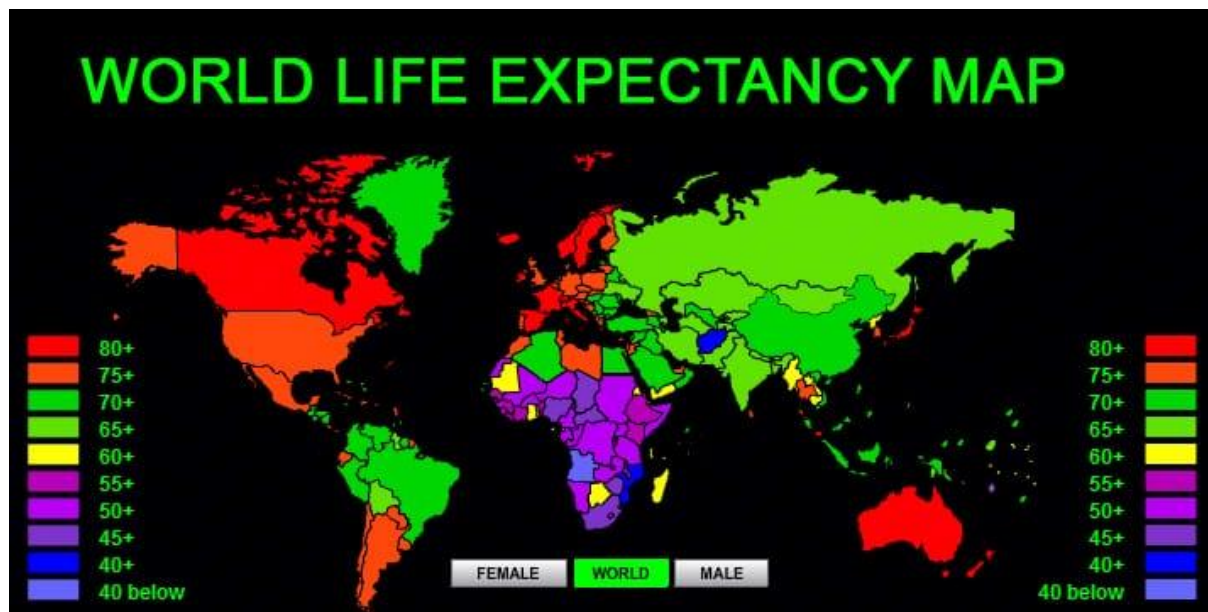
In developed countries, average life expectancy often ranges between 75 and 85 years. These countries have:

- Well-equipped hospitals
- Skilled doctors and medical staff
- High literacy rates
- Strong economies
- Good infrastructure
- Social security systems

In developing countries, life expectancy is usually between 60 and 70 years, and sometimes even lower. These countries often face:

- Poverty and unemployment
- Limited access to healthcare
- Malnutrition
- Poor sanitation and water supply
- High infant and maternal mortality
- Political instability and conflicts

The gap between developed and developing countries clearly shows that life expectancy is closely linked with economic and social development. People in developed nations enjoy longer and healthier lives, while people in developing nations often struggle with basic survival needs.



5. Importance of Life Expectancy

Life expectancy is an important indicator for understanding the quality of life in a society. Governments, researchers, and international organizations use life expectancy data to measure development and plan policies.

The importance of life expectancy includes:

- Measuring national development
- Understanding public health conditions
- Planning healthcare systems
- Identifying social and economic problems
- Comparing countries and regions
- Guiding policy decisions

A high life expectancy indicates that people have access to good healthcare, education, and living conditions. A low life expectancy signals the need for urgent improvements in health, economy, and social systems.

6. Factors Affecting Life Expectancy

Life expectancy is influenced by many factors. These factors can be broadly classified into biological, social, economic, environmental, and lifestyle factors.

(a) Biological Factors

Genetics plays an important role in determining how long a person lives. Some people inherit strong immunity and resistance to diseases, while others may be more vulnerable. Gender also affects life expectancy; generally, women tend to live longer than men.

(b) Healthcare and Medical Factors

The availability and quality of healthcare services significantly affect life expectancy. Countries with advanced medical facilities, vaccination programs, and disease control systems have higher life expectancy. Lack of healthcare leads to higher death rates.

(c) Economic Factors

Income level and economic stability strongly influence life expectancy. Wealthier individuals and nations can afford better healthcare, nutrition, and living conditions. Poverty often leads to poor health and shorter lifespans.

(d) Educational Factors

Education increases awareness about health, hygiene, and nutrition. Educated people are more likely to adopt healthy lifestyles and seek medical help when needed, which increases life expectancy.

(e) Lifestyle Factors

Lifestyle choices such as diet, exercise, smoking, alcohol consumption, and stress levels directly affect life expectancy. Healthy habits increase lifespan, while unhealthy habits reduce it.

(f) Environmental Factors

Environmental conditions such as air pollution, water quality, climate, and housing affect health and longevity. Clean environments promote long life, while polluted environments cause diseases and reduce life expectancy.

(g) Social and Political Factors

Social stability, peace, and effective governance also influence life expectancy. Wars, conflicts, and political instability reduce life expectancy by destroying healthcare systems and living conditions.

7. Steps to Increase Life Expectancy

To improve life expectancy, governments and societies must take several important steps.

(1) Strengthening Healthcare Systems

- Building more hospitals and clinics
- Training healthcare professionals
- Providing affordable medicines and treatments
- Expanding vaccination programs

(2) Improving Nutrition and Food Security

- Ensuring availability of nutritious food
- Promoting balanced diets
- Preventing malnutrition and hunger

(3) Providing Clean Water and Sanitation

- Supplying safe drinking water
- Improving sanitation and waste management
- Promoting hygiene practices

(4) Enhancing Education and Awareness

- Increasing literacy rates
- Conducting health awareness programs
- Educating people about healthy lifestyles

(5) Reducing Poverty and Inequality

- Creating employment opportunities
- Improving income levels

- Providing social welfare programs

(6) Protecting the Environment

- Reducing pollution
- Promoting sustainable development
- Encouraging eco-friendly practices

(7) Promoting Healthy Lifestyles

- Encouraging physical activity
- Reducing smoking and alcohol consumption
- Managing stress and mental health

By implementing these measures, countries can significantly improve the quality of life and increase life expectancy.

8. Significance of Life Expectancy in Human Development

Life expectancy is not just a number; it represents the overall progress of humanity. The increase in life expectancy from ancient times to the present shows how human beings have overcome many challenges through knowledge, science, and cooperation.

However, the unequal distribution of life expectancy across the world highlights the need for global efforts to improve health and living conditions in developing countries. Ensuring longer and healthier lives for all people is one of the major goals of sustainable development.

Thus, life expectancy serves as a mirror of human civilization, reflecting both its achievements and challenges.

9. Real world examples of countries that have taken strong steps to improve life expectancy.

Japan – Preventive Healthcare & Healthy Lifestyle

Life Expectancy: Around 84–85 years (one of the highest in the world)

Steps Taken:

- Universal health insurance system

- Regular health checkups for citizens
- Strong focus on preventive healthcare
- Promotion of balanced traditional diet (fish, vegetables, rice)
- Active lifestyle among elderly population

Result:

Japan has one of the lowest rates of heart disease and obesity, contributing to long and healthy lives.

Cuba – Strong Primary Healthcare System

Life Expectancy: Around 78–80 years (high despite limited economic resources)

Steps Taken:

- Free healthcare for all citizens
- Community-based family doctor system
- Emphasis on prevention rather than expensive treatments
- High vaccination coverage

Result:

Cuba achieved life expectancy similar to many developed countries despite being a developing nation.

Rwanda – Expanding Healthcare Access

Life Expectancy: Increased from about 48 years (2000) to over 69 years today

Steps Taken:

- Community-based health insurance
- Training community health workers
- Strong vaccination and maternal care programs
- Fighting infectious diseases like malaria and HIV

Result:

Rwanda became one of Africa's fastest-improving countries in life expectancy.

Singapore – Clean Environment & Efficient Healthcare

Life Expectancy: Around 83–84 years

Steps Taken:

- High-quality and efficient healthcare system
- Strict public health laws
- Clean environment and strong sanitation system
- Public health education campaigns

Result:

Singapore achieved one of the highest life expectancies in Asia.

Brazil – Public Health Programs & Vaccination

Life Expectancy: Around 75 years (steady improvement)

Steps Taken:

- Universal healthcare system (SUS)
- Family Health Strategy program
- Large-scale vaccination campaigns
- Improved sanitation and poverty reduction programs

Result:

Brazil significantly reduced infant mortality and increased average lifespan.

The below table summarises the life expectancy attributes and their descriptions.

Life Expectancy Attributes

Attribute	Description
Country	Name of the country
Year	Year of observation

Attribute	Description
Status	Developed or Developing country
Adult Mortality	Adult mortality rates of both sexes (probability of dying between ages 15 and 60 per 1000 population)
Infant Deaths	Number of infant deaths per 1000 population
Alcohol	Recorded per capita alcohol consumption (15+ years) in litres of pure alcohol
Percentage Expenditure	Health expenditure as a percentage of GDP per capita (%)
Hepatitis B	Hepatitis B (Hep B) immunization coverage among 1-year-olds (%)
Measles	Number of reported measles cases per 1000 population
BMI	Average Body Mass Index of the total population
Under-Five Deaths	Number of deaths among children under five per 1000 population
Polio	Polio (Pol3) immunization coverage among 1-year-olds (%)
Total Expenditure	General government expenditure on health as a percentage of total government expenditure (%)
Diphtheria	DTP3 (Diphtheria, Tetanus, Pertussis) immunization coverage among 1-year-olds (%)
HIV/AIDS	Deaths per 1,000 live births due to HIV/AIDS (ages 0–4)
GDP	Gross Domestic Product per capita (USD)
Population	Total population of the country

Scope of study

The scope of this study explains what topics are covered in this project about life expectancy.

This study includes:

- The meaning and importance of life expectancy
- Life expectancy in ancient times and modern times
- Comparison between developed and developing countries
- Factors that affect life expectancy (healthcare, income, education, lifestyle, environment)
- Steps taken by different countries to increase life expectancy

The study mainly focuses on health, social, and economic factors that influence how long people live.

Limitation Of The Study

Every study has some limitations. This project also has a few limits:

- It is based on secondary data (books, reports, websites).
- It does not include surveys or personal interviews.
- The comparison between countries is general, not detailed.
- It focuses on national averages, not differences within regions.
- Data may change over time due to new diseases, wars, or economic changes.

LITERATURE REVIEW

After looking into various research work which has taken place by different researchers and authors will derive and describe their method and inference from their research via different data visualization techniques.

-year predictive tool of patients from a recent one or two hospital visits. In recent years, this is one of the most accurate predictive models, providing results up to 5 years. Their work was based upon the ensemble of different Machine Learning models. The Ensemble technique is a Machine Learning method in which various basic models are combined to achieve the final predictive model. This was the reason behind the deployment of the final model with precise implementation because it was an ensemble with 75 individual models. To attain such a level of precision, they need to be specific about patients and need to have access to each individual's EMR data. It performed far better than the previous models.

Kyle J. Foreman et al. (2018), proposed a paper related to life expectancy estimation in 'Global Health Metrics'. In which their prediction results were mostly centered on the GBD, risk factors, and injuries analysis of 195 countries from 1990 to 2016. Consideration of that number of health-related factors was quite astonishing, and the projection of each attribute against the health scenario was shown. They took 79 health drivers into account for the estimation. They used the ARMA (Auto Regressive Moving Average) model for the estimation of life expectancy based on the health drivers. The demographic and educational factors of each country were not mentioned or taken into account in this particular paper. In the ARMA model, the auto-regression analysis and moving average methods are used and these two methods are applied to time-series datasets (well-behaved).

Beeksma et al. (2019), presented this paper in BMC Medical Informatics. It is one of the recent researches in the concerned subject area. They proposed the idea of using supervised machine learning using recurrent neural networks on deceased patients by using their medical records. They approached the task with supervised machine learning. Then, they trained and tested the data on LSTM (Long Short-Term Memory) recurrent neural networks. The LSTM method is a form of RNN (Recurrent Neural Networks). In the RNN method, the output from the calculation is taken as input within the current phase. LSTM method was formulated to beat the matter of long-term dependencies of RNN. It is mostly used in the time-series data due to the lags between the important unknown events in the time series. Their model was based on

non-text and text features of medical records. The first one was the base model (containing non-text features), and another one was the keyword model (containing text features in EMR). The Keyword model proposed a better accuracy of 29% than compared to 20% of the base model. But, some limitations of this model were the data availability and not generalized in predictions.

N. Kerdprasop et al. (2017), in their paper, suggested that there is an association of environmental as well as economical factors to life expectancy. They used the Chi-Square Automatic Interaction Detector (CHAID) method for categorical values and regression on continuous and numerical values. The underlying principle under the CHAID algorithm is a Decision tree. It is a tool used to discover the relationship between the variables. Using this technique, they attempted to relate between economic growth & resources of a country to the life expectancy of its people. They have done the prediction of the life expectancy of people living near the Mekong River. Eventually, their results revealed a strong relationship between GDP growth & environment to the life expectancy of people.

Yang, J. (2016), in their paper, presented a model to make basic forecasting for sex-specific and socio-economic-specific Life Expectancy in the Netherlands using a set of critical assumptions about the future trends. They were using the traditional Li-Lee model for the estimation. This Li-Lee model is an extension of the original Lee-Carter model. The Lee-Carter model was introduced in 1992, and as of today, it is used by many authors to determine future mortality trends. This model is fed with details of age-specific mortality rates and the result is another matrix of forecasted mortality rates.

Creating life expectancy forecasting models may be a tedious job to try and do. There are several challenging factors involved in this process. From identification of indicators, collecting the datasets, to preprocess the available data, and at last final phase of implementation by applying appropriate machine learning algorithms on to the datasets. Testing the accuracy is another matter of discussion, but still, it is one of the involved challenges in the process. Now, the authors will discuss each of the difficulties briefly below one by one

1) The first significant challenge in front of us will be to prepare the correct and error-free dataset. The sources of data needed to be legit and trustworthy.

2) After identification of legitimacy of sources, dataset need to be categorized the two categories;

a) Structured data: This kind of data is stored in EMR or UN official records as well as on the government's official websites for public reference.

b) Unstructured data: It consists of numerous non- validated information that needs to be converted into measurable and statistically discrete attributes to accommodate the other characteristics of the dataset.

3) Usage of numerous non-standardized abbreviations and improper attributes (with no proper definition in the context of life expectancy) can create ambiguity in the final results. Elimination of redundant information is also an essential process in the initial stages and also a challenging one.

MOTIVATION

Life expectancy is a key indicator of human development and reflects the overall economic, social, and health conditions of a country. Understanding the factors that influence life expectancy is essential for evaluating the quality of life and identifying inequalities across nations. Despite improvements in global health outcomes, significant gaps in life expectancy continue to exist between developed and developing countries, highlighting persistent disparities in resources, healthcare systems, and socio-economic conditions.

The motivation for this study arises from the need to systematically analyze these differences using statistical methods. By applying one-way ANOVA, correlation analysis, and linear and multiple regression models, this research seeks to quantify the extent of variation in life expectancy between developed and developing countries and to identify the key determinants responsible for these differences. Such an analysis not only enhances the understanding of global health inequalities but also provides empirical evidence that can support policy formulation and development strategies aimed at improving life expectancy in developing nations.

Furthermore, this study is motivated by the growing importance of data-driven decision-making in public health and economic planning. By integrating economic, social, and health variables into a comprehensive analytical framework, the research contributes to a deeper understanding of how multiple factors interact to influence life expectancy across different levels of national development.

PROBLEM DEFINITION

Life expectancy is widely recognized as a comprehensive indicator of a nation's health status and socio-economic development. Despite global advancements in healthcare and living standards, substantial disparities in life expectancy persist between developed and developing countries. These disparities are influenced by a complex interplay of economic, social, and health-related factors, including income levels, educational attainment, healthcare expenditure, and mortality patterns.

However, existing studies often examine these factors in isolation and do not sufficiently quantify the extent to which life expectancy differs between developed and developing countries using rigorous statistical approaches. Furthermore, there is limited empirical evidence on how strongly life expectancy is associated with key socio-economic and health indicators and which of these factors serve as significant predictors across different levels of development.

Therefore, the central problem addressed in this study is to systematically investigate the differences in life expectancy between developed and developing countries and to analyse the relationships and predictive effects of economic, social, and health variables. By applying one-way ANOVA, two-way ANOVA, correlation analysis, multiple regression model, this study aims to provide a comprehensive statistical understanding of the determinants of life expectancy and the nature of developmental disparities between countries.

OBJECTIVES

- To examine the trend in life expectancy during the period 2016-2024
- To test the significance difference between developed and developing countries.
- To examine the relationship between dependent variable and independent variables.
- To predict life expectancy after a 10 year period using multiple regression.
- To evaluate the performance of models using R^2 and RMSE values

METHODOLOGY

Research Design

This study adopts a quantitative and explanatory research design to investigate the factors influencing life expectancy and to examine the differences between developed and developing countries. Quantitative methods are appropriate because the study relies on numerical data and statistical techniques to analyse relationships and differences among variables. The research focuses on identifying economic, social, and health determinants of life expectancy and assessing whether these determinants vary significantly between developed and developing countries.

Data Source and Dataset Description

The study uses secondary data collected from international databases such as the World Health Organization (WHO), World Bank, and United Nations Development Programme (UNDP). The dataset covers multiple countries over the period 2016 to 2024 and includes variables related to life expectancy, economic indicators, social factors, and health conditions.

Each observation in the dataset represents a country-year combination. Countries are classified into two groups based on their level of development: developed and developing countries. This classification is used as the basis for comparative analysis throughout the study.

The dataset contains the following key variables:

Life Expectancy (years)

Economic indicators: GDP, total expenditure, percentage expenditure, income composition of resources

Social indicators: schooling, alcohol consumption

Health indicators: adult mortality, infant deaths, under-five deaths, immunization rates (Hepatitis B, Polio, Diphtheria, Measles), HIV/AIDS prevalence, BMI, and thinness indicators

Demographic indicators: population, year, and country status (developed or developing)

Life expectancy is treated as the dependent variable, while economic, social, and health indicators are treated as independent variables.

Classification of Variables

For systematic analysis, the variables are grouped into categories based on their nature and theoretical relevance.

Dependent Variable

The dependent variable in this study is life expectancy, which represents the average number of years a person is expected to live in a given country.

Independent Variables

(a) **Economic Variables:** Economic variables reflect the financial and developmental capacity of countries. These include GDP, total expenditure, percentage expenditure, and income composition of resources.

(b) Social Variables

Social variables represent human development and lifestyle factors. These include schooling and alcohol consumption.

(c) Health Variables

Health variables directly influence survival rates and population health. These include adult mortality, infant deaths, under-five deaths, immunization indicators, HIV/AIDS prevalence, BMI, and thinness indicators.

(d) Demographic Variables

Demographic variables describe population characteristics and include population size, year, country, and development status.

This classification enables a comprehensive understanding of how different dimensions of development influence life expectancy.

Data Preparation and Preprocessing

Before statistical analysis, the dataset was carefully prepared and cleaned. Missing values were identified and handled using appropriate methods, such as removal of incomplete observations or imputation where necessary. Outliers were examined to ensure that extreme values did not distort the results. Variables were checked for consistency and accuracy.

Countries were categorized into developed and developing groups based on established international classifications. Descriptive statistics, including mean, standard deviation,

minimum, and maximum values, were calculated to understand the distribution and basic characteristics of the data.

Analytical Framework

To achieve the objectives of the study, multiple statistical techniques were applied, including correlation analysis, one-way ANOVA, simple linear regression, and multiple regression analysis. These methods were chosen because they allow both comparison of groups and examination of relationships among variables.

Trend Analysis

Trend analysis was carried out to examine the change in life expectancy from 2016 to 2024. A linear regression model was applied with Year as the independent variable and Life Expectancy as the dependent variable to measure the annual rate of change.

Separate regression models were estimated for developed and developing countries to compare their growth patterns. An interaction model ($\text{Year} \times \text{Status}$) was also used to test whether the rate of increase differs significantly between the two groups.

Correlation Analysis

Correlation analysis was used to examine the relationship between life expectancy and various economic, social, and health indicators. Pearson's correlation coefficient was employed to measure the strength and direction of linear relationships between variables.

Correlation analysis was conducted separately for developed and developing countries to identify differences in the patterns of relationships between the two groups.

The correlation coefficient ranges from -1 to +1, where values close to +1 indicate a strong positive relationship, values close to -1 indicate a strong negative relationship, and values close to 0 indicate no relationship. Statistical significance was evaluated using p-values at a 5% significance level.

The purpose of correlation analysis in this study is to identify key factors associated with life expectancy and to compare the nature of these associations between developed and developing countries.

One-Way ANOVA (Analysis of Variance)

One-way Analysis of Variance (ANOVA) was applied to test whether there is a statistically significant difference in life expectancy between developed and developing countries.

Rationale for Using One-Way ANOVA

One-way ANOVA is appropriate when comparing the means of a continuous variable across two or more independent groups. In this study, the independent variable is country development status with two categories (developed and developing), and the dependent variable is life expectancy.

Hypotheses

The hypotheses for ANOVA are formulated as follows:

Null hypothesis (H_0): There is no significant difference in mean life expectancy between developed and developing countries.

Alternative hypothesis (H_1): There is a significant difference in mean life expectancy between developed and developing countries.

Procedure

The ANOVA procedure involves partitioning the total variation in life expectancy into two components: variation between groups (developed vs developing countries) and variation within groups (countries within each category). The F-statistic is calculated as the ratio of between-group variance to within-group variance.

The decision rule is based on the p-value. If the p-value is less than 0.05, the null hypothesis is rejected, indicating a significant difference between developed and developing countries.

Two-Way ANOVA (Two-Way Classification)

In addition to one-way ANOVA, **Two-Way ANOVA** was applied to examine the combined effect of two categorical factors on life expectancy.

Factors Considered:

- Factor A: Country Classification (Developed / Developing)

- Factor B: Year (or another categorical grouping variable)

Two-way ANOVA allows us to analyse:

1. The main effect of country classification
2. The main effect of the second factor (e.g., year)
3. The interaction effect between country classification and the second factor

Hypotheses for Two-Way ANOVA:

1 For Country Classification

- H_0 : No significant difference in life expectancy between developed and developing countries
- H_1 : Significant difference exists

2 For Second Factor (e.g., Year)

- H_0 : No significant difference across years
- H_1 : Significant difference across years

3 For Interaction Effect

- H_0 : No interaction between country type and year
- H_1 : There is an interaction effect

This method helps determine whether the difference in life expectancy depends not only on development status but also on time (or another classification factor).

Regression Analysis

Regression analysis was used to examine the impact of independent variables on life expectancy and to develop predictive models.

Multiple Regression Analysis

Multiple regression analysis was conducted to assess the combined effect of economic, social, and health variables on life expectancy.

The multiple regression model is expressed as:

$$\text{Life Expectancy} = \beta_0 + \beta_1(\text{GDP}) + \beta_2(\text{Schooling}) + \beta_3(\text{Health Indicators}) + \beta_4(\text{Other Variables}) + \varepsilon$$

This model allows the study to identify the relative importance of different determinants of life expectancy while controlling for other factors.

Separate multiple regression models were estimated for developed and developing countries to compare the determinants of life expectancy in the two groups.

Model Evaluation and Interpretation

The regression models were evaluated using statistical indicators such as R^2 (coefficient of determination), adjusted R^2 , F-statistics, and p-values. These measures indicate how well the independent variables explain variations in life expectancy.

Regression coefficients were interpreted to understand the direction and magnitude of the effect of each independent variable on life expectancy.

RESULTS AND ANALYSIS

SOURCE: [Country's life expectancy data\(2016-2024\)](#)

Trend Analysis

Program

```
trend_model <- lm(Life.Expectancy ~ Year, data = data)
```

```
> summary(trend_model)
```

Call:

```
lm(formula = Life.Expectancy ~ Year, data = data)
```

Residuals:

Min	1Q	Median	3Q	Max
-20.575	-5.585	1.517	5.504	13.825

Coefficients:

Term	Estimate	Std. Error	t value	Pr(> t)	Significance
(Intercept)	-730.0021	149.2808	-4.890	1.11e-06	***
Year	0.3973	0.0739	5.376	8.69e-08	***

Residual standard error: 7.744 on 1645 degrees of freedom

Multiple R-squared: 0.01727, Adjusted R-squared: 0.01667

F-statistic: 28.91 on 1 and 1645 DF, p-value: 8.691e-08

```
print(trend_data)
```

```
# A tibble: 18 × 3
```

```
# Groups: Year [9]
```

Year	Status	Mean_Life_Expectancy
1 2016	Developed	79.5
2 2016	Developing	68.5
3 2017	Developed	80.0
4 2017	Developing	68.9
5 2018	Developed	80.4
6 2018	Developing	69.3
7 2019	Developed	80.9
8 2019	Developing	69.7
9 2020	Developed	81.1
10 2020	Developing	70.1
11 2021	Developed	81.6
12 2021	Developing	70.5
13 2022	Developed	82.0
14 2022	Developing	70.9
15 2023	Developed	82.5
16 2023	Developing	71.3
17 2024	Developed	82.5
18 2024	Developing	71.6

```
>library(ggplot2)
```

```
>
```

```
> ggplot(trend_data, aes(x = Year,
```

```
+           y = Mean_Life_Expectancy,
```

```
+           color = Status)) +
```



```

+ geom_line(size = 1.2) +
+ geom_point() +
+ labs(title = "Trend of Life Expectancy (2016–2024)",
+       x = "Year",
+       y = "Average Life Expectancy") +
+ theme_minimal()
> trend_dev <- lm(Life.Expectancy ~ Year,
+                 data = subset(data, Status == "Developed"))
> summary(trend_dev)

```

Call:

```
lm(formula = Life.Expectancy ~ Year, data = subset(data, Status ==
"Developed"))
```

Residuals:

min	1Q	Median	3Q	Max
-6.1547	-2.8612	0.7127	2.4997	5.2453

Coefficients:

Term	Estimate	Std. Error	t value	Pr(> t)	Significance
(Intercept)	-713.68103	123.24057	-5.791	1.50e-08	***
Year	0.39350	0.06101	6.450	3.54e-10	***

Residual standard error: 3.026 on 367 degrees of freedom

Multiple R-squared: 0.1018, Adjusted R-squared: 0.09936

F-statistic: 41.6 on 1 and 367 DF, p-value: 3.539e-10

```
> trend_devg <- lm(Life.Expectancy ~ Year,
+ data = subset(data, Status == "Developing"))
> summary(trend_devg)
```

Call:

```
lm(formula = Life.Expectancy ~ Year, data = subset(data, Status ==
"Developing"))
```

Residuals:

Min	1Q	Median	3Q	Max
-18.097	-5.002	1.693	5.693	12.303

Coefficients:

Term	Estimate	Std. Error	t value	Pr(> t)	Significance
(Intercept)	-734.71452	150.25521	-4.890	1.14e-06	***
Year	0.39842	0.07438	5.356	1.01e-07	***

Residual standard error: 6.866 on 1276 degrees of freedom

Multiple R-squared: 0.02199, Adjusted R-squared: 0.02122

F-statistic: 28.69 on 1 and 1276 DF, p-value: 1.006e-07

```
> trend_interaction <- lm(Life.Expectancy ~ Year * Status, data = data)
> summary(trend_interaction)
```

Call:

```
lm(formula = Life.Expectancy ~ Year * Status, data = data)
```

Residuals:

Min	1Q	Median	3Q	Max
-18.097	-3.989	1.295	4.595	12.303

Coefficients:

Term	Estimate	Std. Error	t value	Pr(> t)	Significance
(Intercept)	-7.137e+02	2.532e+02	-2.818	0.00488	**
Year	3.935e-01	1.254e-01	3.139	0.00173	**
StatusDeveloping	-2.103e+01	2.875e+02	-0.073	0.94168	
Year:StatusDeveloping	4.929e-03	1.423e-01	0.035	0.97237	

Residual standard error: 6.217 on 1643 degrees of freedom

Multiple R-squared: 0.3673, Adjusted R-squared: 0.3661

F-statistic: 317.9 on 3 and 1643 DF, p-value: < 2.2e-16

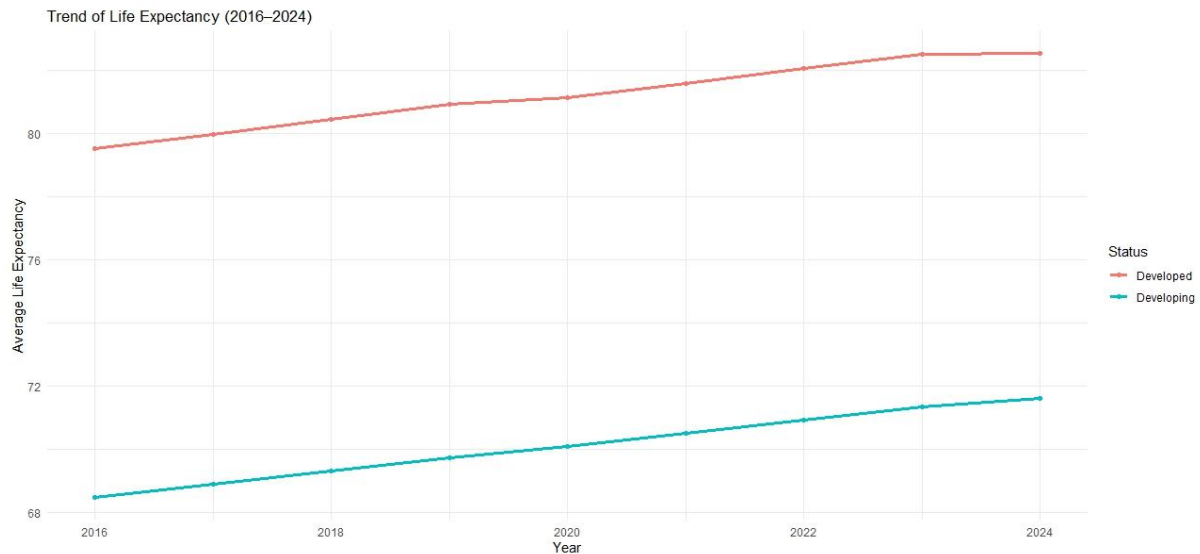
```
> trend_summary <- trend_data %>%
+   group_by(Status) %>%
+   summarise(
+     First_Year = first(Mean_Life_Expectancy),
+     Last_Year = last(Mean_Life_Expectancy),
+     Total_Growth = Last_Year - First_Year
+   )
>
```

```
> print(trend_summary)
```

```
# A tibble: 2 × 4
```

Status	First_Year	Last_Year	Total_Growth
1 Developed	79.5	82.5	3.03
2 Developing	68.5	71.6	3.15

Output



The above life expectancy trend analysis was conducted to examine changes in life expectancy from **2016 to 2024**. The regression results show a **significant positive trend** ($\beta = 0.3973$, $p < 0.001$), indicating that life expectancy increased gradually over time.

When analyzed separately, both **developed and developing countries** showed a significant upward trend in life expectancy. Developed countries increased from **79.5 years in 2016 to 82.5 years in 2024**, while developing countries increased from **68.5 years to 71.6 years** during the same period.

Although both groups experienced improvements, **developed countries consistently maintained higher life expectancy levels than developing countries**, indicating a persistent gap between the two groups.

ANOVA

One-Way ANOVA

Program

```
> anova_model <- aov(Life.Expectancy ~ Status, data = data)
```

```
> summary(anova_model)
```

Output:

Source	Df	Sum Sq	Mean Sq	F value	Pr(>F)	Significance
Status	1	35133	35133	885.8	<2e-16	***
Residuals	1645	65245	40	—	—	—

Commands:

```
# Normality of residuals
```

```
> shapiro.test(residuals(anova_model))
```

Output:

Shapiro-Wilk normality test

data: residuals(anova_model)

W = 0.97083, p-value < 2.2e-16

Commands:

```
# Homogeneity of variance
```

```
> bartlett.test(Life.Expectancy ~ Status, data = data)
```

Output:

Bartlett test of homogeneity of variances

data: Life.Expectancy by Status

Bartlett's K-squared = 252.75, df = 1, p-value < 2.2e-16

Commands:

```
> t_test_result <- t.test(Life.Expectancy ~ Status,
```

```

+           data = data,
+           var.equal = FALSE)
> t_test_resultt_test_result <- t.test(Life.Expectancy ~ Status,
+           data = data,
+           var.equal = FALSE)
>
> t_test_result

```

output

Welch Two Sample t-test

data: Life.Expectancy by Status

t = 43.368, df = 1340.5, p-value < 2.2e-16

alternative hypothesis: true difference in means between group Developed and group Developing is not equal to 0

95 percent confidence interval:

10.57596 11.57808

sample estimates:

mean in group Developed mean in group Developing

81.18076 70.10374

Commands:

```

> t.test(Life.Expectancy ~ Status,
+   data = data,
+   var.equal = TRUE)

```

Output:

Two Sample t-test

data: Life.Expectancy by Status

t = 29.762, df = 1645, p-value < 2.2e-16

alternative hypothesis: true difference in means between group Developed and group Developing is not equal to 0

95 percent confidence interval:

10.34701 11.80703

sample estimates:

mean in group Developed mean in group Developing

81.18076 70.103

A one-way ANOVA was conducted to examine whether life expectancy differs between **developed and developing countries**. The results show a **statistically significant difference** in life expectancy between the two groups ($F = 885.8$, $p < 0.001$).

The average life expectancy in **developed countries (81.18 years)** is considerably higher than in **developing countries (70.10 years)**. The t-test results also confirm this significant difference ($t = 43.37$, $p < 0.001$).

These findings indicate that **country development status has a strong influence on life expectancy**, with developed countries experiencing higher average life expectancy compared to developing countries.

Two-way ANOVA

Program

```
two_way_anova <- aov(Life.Expectancy ~ Status * Year, data = data)
```

```
> summary(two_way_anova)
```

Output:

Source	Df	Sum Sq	Mean Sq	F value	Pr(>F)	Significance
Status	1	35133	35133	901.194	< 2e-16	***
Year	8	1738	217	5.573	5.52e-07	***
Status:Year	8	1	0.005	0.005	1	(none)

Source	Df	Sum Sq	Mean Sq	F value	Pr(>F)	Significance
--------	----	--------	---------	---------	--------	--------------

Residuals	1629	63506	39	—	—	—
-----------	------	-------	----	---	---	---

Commands:

```
> shapiro.test(residuals(two_way_anova))
```

Output:

Shapiro-Wilk normality test

data: residuals(two_way_anova)

W = 0.96712, p-value < 2.2e-16

Commands:

```
> bartlett.test(Life.Expectancy ~ interaction(Status, Year), data = data)
```

Output:

Bartlett test of homogeneity of variances

data: Life.Expectancy by interaction(Status, Year)

Bartlett's K-squared = 266.97, df = 17, p-value < 2.2e

A two-way ANOVA was conducted to examine the effect of **country status and year on life expectancy**. The results show that **status has a significant effect on life expectancy** ($F = 901.19$, $p < 0.001$), indicating that developed countries have higher life expectancy than developing countries.

The **year factor is also statistically significant** ($F = 5.57$, $p < 0.001$), suggesting that life expectancy has changed over time. However, the **interaction between status and year is not significant** ($F = 0.005$, $p = 1$), indicating that the trend over time is similar for both developed and developing countries.

CORRELATION

Program

```
cor_life_dev <- cor_dev["Life.Expectancy", ]  
  
cor_life_dev_df <- data.frame(  
  Variable = names(cor_life_dev),  
  Correlation_with_Life_Expectancy = round(as.numeric(cor_life_dev), 3)  
)
```

Output

cor_life_dev_df

Variable	Correlation_with_Life_Expectancy
Year	0.319
Life.Expectancy	1
Adult.Mortality	-0.271
Infant.Deaths	-0.575
Alcohol	0.016
Percentage.Expenditure	0.129
Hepatitis.B	-0.038
Measles	-0.298
BMI	-0.006
Under.Five.Deaths	-0.472
Polio	NA
Total.Expenditure	-0.196
Diphtheria	NA
HIV.AIDS	-0.124
GDP	-0.01
Population	0.018
Thinness.1.19.Years	-0.065
Thinness.5.9.Years	-0.065
Income.Composition.of.Resources	0.052
Schooling	0.043

Developing countries

Program

```
cor_life_devg <- cor_devg["Life.Expectancy", ]  
  
cor_life_devg_df <- data.frame(  
  Variable = names(cor_life_devg),  
  Correlation_with_Life_Expectancy = round(as.numeric(cor_life_devg), 3)  
)  
  
cor_life_devg_df
```

output

Variable	Correlation_with_Life_Expectancy
Year	0.148
Life.Expectancy	1
Adult.Mortality	-0.835
Infant.Deaths	-0.845
Alcohol	-0.123
Percentage.Expenditure	0.514
Hepatitis.B	-0.45
Measles	-0.109
BMI	0.661
Under.Five.Deaths	-0.801
Polio	-0.244
Total.Expenditure	0.216
Diphtheria	-0.244
HIV.AIDS	-0.603
GDP	0.211
Population	0.041
Thinness.1.19.Years	-0.6
Thinness.5.9.Years	-0.538
Income.Composition.of.Resources	0.294
Schooling	0.408

Correlation compartion

Program

```
cor_life_dev_df <- cor_life_dev_df %>%  
  
rename(Correlation_Developed = Correlation_with_Life_Expectancy)  
  
cor_life_devg_df <- cor_life_devg_df %>%  
  
rename(Correlation_Developing = Correlation_with_Life_Expectancy)  
  
cor_comparison <- merge(  
  
cor_life_dev_df,  
  
cor_life_devg_df,  
  
by = "Variable",  
  
all = TRUE  
)
```

Output

```
cor_comparison
```

Variable	Correlation_Developed	Correlation_Developing
Life.Expectancy	1	1
Infant.Deaths	-0.575	-0.845
Adult.Mortality	-0.271	-0.835
Under.Five.Deaths	-0.472	-0.801
BMI	-0.006	0.661
HIV.AIDS	-0.124	-0.603
Thinness.1.19.Years	-0.065	-0.6
Thinness.5.9.Years	-0.065	-0.538
Percentage.Expenditure	0.129	0.514
Hepatitis.B	-0.038	-0.45
Schooling	0.043	0.408
Income.Composition.of.Resources	0.052	0.294
Diphtheria	NA	-0.244
Polio	NA	-0.244

Total.Expenditure	-0.196	0.216
GDP	-0.01	0.211
Year	0.319	0.148
Alcohol	0.016	-0.123
Measles	-0.298	-0.109
Population	0.018	0.041

Correlation analysis was conducted to examine the relationship between life expectancy and various health and socio-economic factors in developed and developing countries.

In **developed countries**, life expectancy shows moderate negative correlations with **infant deaths (-0.575)**, **under-five deaths (-0.472)**, and **adult mortality (-0.271)**, indicating that higher mortality rates are associated with lower life expectancy. Most other variables show weak correlations, suggesting that life expectancy in developed countries is relatively stable and less influenced by individual factors.

In **developing countries**, stronger relationships are observed. Life expectancy has strong negative correlations with **infant deaths (-0.845)**, **adult mortality (-0.835)**, **under-five deaths (-0.801)**, and **HIV/AIDS (-0.603)**. Positive correlations are observed with **BMI (0.661)**, **health expenditure (0.514)**, and **schooling (0.408)**, indicating that improvements in health conditions, education, and healthcare spending are associated with higher life expectancy.

Overall, the results suggest that **health, nutrition, and healthcare-related factors have a stronger impact on life expectancy in developing countries compared to developed countries.**

MULTIPLE LINEAR REGRESSION

Developed countries

Program

```
num_cols <-
c("Year", "Life.Expectancy", "GDP", "Population", "Schooling", "BMI", "Adult.Mortality")

data[num_cols] <- lapply(data[num_cols], function(x) {
  x <- gsub(",", "", x)
```

```

as.numeric(x)

})

data_clean <- data[!is.na(data$Life.Expectancy), ]

for(col in c("GDP", "Population", "Schooling", "BMI", "Adult.Mortality")){

data_clean[[col]][is.na(data_clean[[col]])] <- mean(data_clean[[col]], na.rm = TRUE)

}

train_data <- data_clean[data_clean$Year <= 2024, ]

model_global <- lm(Life.Expectancy ~ Year + GDP + Population + Schooling + BMI +
Adult.Mortality,
data = train_data)

summary(model_global)

life_2024_global <- mean(train_data$Life.Expectancy[train_data$Year == 2024], na.rm =
TRUE)

life_2024_global

future_global <- data.frame(

Year = 2034,

GDP = mean(train_data$GDP, na.rm = TRUE),

Population = mean(train_data$Population, na.rm = TRUE),

Schooling = mean(train_data$Schooling, na.rm = TRUE),

BMI = mean(train_data$BMI, na.rm = TRUE),

Adult.Mortality = mean(train_data$Adult.Mortality, na.rm = TRUE)

)

life_2034_global <- predict(model_global, newdata = future_global)

life_2034_global

diff_global <- life_2034_global - life_2024_global

diff_global

output_global <- data.frame(

```

```
Year = c("2024 (Actual)", "2034 (Predicted)", "Difference (10 Years)"),
```

```
Life_Expectancy = c(life_2024_global, life_2034_global, diff_global)
```

```
)
```

```
output_global
```

```
countries <- unique(data_clean$Country)
```

```
results_country <- data.frame()
```

```
output
```

Country	Life_2024	Life_2034	Difference_10_Years
Australia	83.6	85.6	2
Austria	82.6	84.6	2
Belgium	82.6	84.6	2
Canada	83.1	85.1	2
Denmark	82.1	84.1	2
Estonia	81	86	5
Finland	84	89	5
France	86	91	5
Germany	85	90	5
Greece	84	89	5
Hungary	78	77.77777778	-0.22222222
Iceland	84	81.44444444	-2.55555556
Ireland	88	84.88888889	-3.11111111
Israel	88	84.88888889	-3.11111111
Italy	88	84.88888889	-3.11111111
Japan	85.5	84.56666667	-0.93333333
Kuwait	82	87	5
Latvia	77	74.44444444	-2.55555556
Lebanon	80	81.68686869	1.68686869
Lithuania	77.5	75.94444444	-1.55555556
Luxembourg	85	83.44444444	-1.55555556
Malta	85	83.44444444	-1.55555556

Mauritius	79	84	5
Montenegro	81	79.44444444	-1.555555556
Netherlands	83	80.44444444	-2.555555556
New Zealand	84	76.77777778	-7.222222222
Norway	85	77.77777778	-7.222222222
Poland	80	85	5
Portugal	82	87	5
Republic of Korea	85.5	90.5	5
Romania	78	83	5
Saudi Arabia	76.6	78.6	2
Seychelles	79	84	5
Singapore	87	92	5
Slovakia	77	76.77777778	-0.222222222
Slovenia	78	75.44444444	-2.555555556
Spain	87	92	5
Sweden	84	89	5
Switzerland	86	91	5
United Kingdom of Great Britain and Northern Ireland	82.1	84.1	2
United States of America	77.9	76.9	-1

DEVELOPING COUNTRIES

Program

```

for(ct in countries){

country_data <- data_clean[data_clean$Country == ct & data_clean$Year <= 2024, ]

if(nrow(country_data) < 5) next # skip countries with very little data

model_country <- lm(Life.Expectancy ~ Year + GDP + Population + Schooling + BMI +
Adult.Mortality,

data = country_data)

# Actual 2024 life expectancy

life_2024 <- mean(country_data$Life.Expectancy[country_data$Year == 2024], na.rm =
TRUE)

```

```

# Predict 2034

future_country <- data.frame(

Year = 2034,

GDP = mean(country_data$GDP, na.rm = TRUE),

Population = mean(country_data$Population, na.rm = TRUE),

Schooling = mean(country_data$Schooling, na.rm = TRUE),

BMI = mean(country_data$BMI, na.rm = TRUE),

Adult.Mortality = mean(country_data$Adult.Mortality, na.rm = TRUE)

)

life_2034 <- predict(model_country, newdata = future_country)

diff <- life_2034 - life_2024

results_country <- rbind(results_country, data.frame(

Country = ct,

Life_2024 = life_2024,

Life_2034 = life_2034,

Difference_10_Years = diff

))

}

results_dev_country

```

output:

Country	Life_2024	Life_2034	Difference_10_Years
Afghanistan	60.5	59.88137	-0.618629325
Albania	80.1	82.1	2
Algeria	78.1	80.1	2
Angola	62.9	65.9	3
Antigua and Barbuda	76.6	78.6	2

Argentina	77.1	79.1	2
Armenia	76.1	78.1	2
Azerbaijan	73.1	75.1	2
Bahamas	76.6	78.6	2
Bahrain	76.6	78.6	2
Bangladesh	73.6	75.6	2
Barbados	76.6	78.6	2
Belarus	73.6	75.6	2
Belize	75.1	77.1	2
Benin	62.6	64.6	2
Bhutan	70.6	72.6	2
Bolivia	70.6	72.6	2
Bosnia and Herzegovina	77.6	79.6	2
Botswana	64.6	66.6	2
Brazil	74.3	75.3	1
Brunei Darussalam	76.5	74.01111	-2.488888889
Bulgaria	76.1	78.1	2
Burkina Faso	60.6	62.6	2
Burundi	61.6	63.6	2
Côte d'Ivoire	60.6	62.6	2
Cabo Verde	73.6	75.6	2
Cambodia	71.6	73.6	2
Cameroon	59.6	61.6	2
Central African Republic	53.6	55.6	2
Chad	54.6	56.6	2
Chile	80.6	82.6	2
China	77.6	79.6	2
Colombia	76.6	78.6	2
Comoros	64.6	66.6	2
Congo	53.6	55.6	2
Costa Rica	80.6	82.6	2
Croatia	79.6	81.6	2

Cuba	80.6	82.6	2
Cyprus	80.1	82.1	2
Czechia	79.1	81.1	2
Democratic People's Republic of Korea	72.6	74.6	2
Democratic Republic of the Congo	60.6	62.6	2
Djibouti	62.6	64.6	2
Dominican Republic	75.6	77.6	2
Ecuador	76.6	78.6	2
Egypt	74.1	76.1	2
El Salvador	72.6	74.6	2
Equatorial Guinea	66.6	68.6	2
Eritrea	66.6	68.6	2
Ethiopia	73	83	10
Fiji	76	81	5
Gabon	73	83	10
Gambia	71	81	10
Georgia	78	83	5
Ghana	71	81	10
Grenada	78	83	5
Guatemala	76	81	5
Guinea	68	78	10
Guinea-Bissau	71	81	10
Guyana	76	86	10
Haiti	66	67.22222	1.22222222
Honduras	75	76.22222	1.22222222
India	70	82.88889	12.88888889
Indonesia	73	76.08889	3.088888889
Iran (Islamic Republic of)	80	76.88889	-3.11111111
Iraq	74	70.88889	-3.11111111
Jamaica	77.2	75.6	-1.6
Jordan	76.5	74.94444	-1.555555556
Kazakhstan	80	90	10

Kenya	76	86	10
Kiribati	73	83	10
Kyrgyzstan	78	88	10
Lao People's Democratic Republic	71	73.38889	2.388888889
Lesotho	63.5	58.79024	-4.70976099
Liberia	62	67	5
Libya	74	72.44444	-1.555555556
Madagascar	67	72	5
Malawi	64	74	10
Malaysia	83	93	10
Maldives	84	94	10
Mali	67	77	10
Mauritania	66	64.44444	-1.555555556
Mexico	81.5	86.5	5
Micronesia (Federated States of)	69	74	5
Mongolia	71	69.44444	-1.555555556
Morocco	76	74.44444	-1.555555556
Mozambique	59	57.44444	-1.555555556
Myanmar	69	67.44444	-1.555555556
Namibia	68	70.38889	2.388888889
Nepal	73	85.88889	12.88888889
Nicaragua	77	75.88889	-1.111111111
Niger	63	75.88889	12.88888889
Nigeria	58	70.88889	12.88888889
Oman	80	78.88889	-1.111111111
Pakistan	71	76	5
Panama	83	88	5
Papua New Guinea	69	74	5
Paraguay	79	84	5
Peru	79	84	5
Philippines	68.5	72.81556	4.315555556
Qatar	81	86	5

Republic of Moldova	74	79	5
Russian Federation	76	81	5
Rwanda	69	74	5
Saint Lucia	77.3	-119.822	-197.1222222
Saint Vincent and the Grenadines	78.3	77.17778	-1.122222222
Samoa	74.3	77.21444	2.914444444
Sao Tome and Principe	70.3	69.17778	-1.122222222
Senegal	66	71	5
Serbia	79	84	5
Sierra Leone	56	61	5
Solomon Islands	71	69.88889	-1.111111111
Somalia	58	56.88889	-1.111111111
South Africa	66	67.22222	1.222222222
South Sudan	61	66	5
Sri Lanka	79	84	5
Sudan	69	74	5
Suriname	76	81	5
Swaziland	58	56.88889	-1.111111111
Syrian Arab Republic	68	69.22222	1.222222222
Tajikistan	68	70.38889	2.388888889
Thailand	73	74.22222	1.222222222
The former Yugoslav republic of Macedonia	73	75.38889	2.388888889
Timor-Leste	68	69.22222	1.222222222
Togo	66	68.38889	2.388888889
Tonga	74	79	5
Trinidad and Tobago	77	82	5
Tunisia	80	85	5
Turkey	81	86	5
Turkmenistan	76	81	5
Uganda	63	64.22222	1.222222222
Ukraine	73.1	74.12222	1.022222222

United Arab Emirates	79	84	5
United Republic of Tanzania	67	72	5
Uruguay	78.1	80.1	2
Uzbekistan	74.5	79.5	5
Vanuatu	72	77	5
Venezuela (Bolivarian Republic of)	66.5	61.5	-5
Viet Nam	75.9	78.9	3
Yemen	60.5	55.5	-5
Zambia	61.5	66.5	5
Zimbabwe	64	69	5

Multiple linear regression analysis was conducted to predict life expectancy using several explanatory variables, including year, GDP, population, schooling, BMI, and adult mortality. The model was used to estimate life expectancy for the year 2034 and compare it with the actual values in 2024.

For **developed countries**, the results show that life expectancy is expected to increase slightly for many countries over the next ten years. Several countries such as France, Germany, Finland, Spain, and Singapore show an increase of about **5 years**, while some countries show only small improvements or slight decreases. This indicates that life expectancy in developed countries is already high and changes over time are relatively small.

For **developing countries**, the predicted changes are more varied. Many countries such as **India, Nepal, Niger, Nigeria, and Ethiopia** show large increases in life expectancy over the next decade. This suggests that improvements in economic development, healthcare access, education, and living conditions could significantly increase life expectancy in these countries.

Overall, the regression analysis indicates that **socio-economic factors such as education, health conditions, and economic development play an important role in determining life expectancy**, especially in developing countries where greater improvements are possible.

MODEL ACCURACY

MULTIPLE LINEAR REGRESSION

Program

```
cat("Developed Countries multiple Model:\n")

cat("R-square =", R2_developed, "\n")

cat("RMSE =", rmse_developed, "\n")

cat("Developing Countries multiple Model:\n")

cat("R-square =", R2_developing, "\n")

cat("RMSE =", rmse_developing, "\n")

model_comparison <- data.frame(

Model = c("Developed Countries", "Developing Countries"),

R_Square = c(R2_developed, R2_developing),

RMSE = c(rmse_developed, rmse_developing)

)
```

Output

Model	R_Square	RMSE
Developed Countries	0.188707642	2.868093844
Developing Countries	0.778343848	3.266072486

The accuracy of the multiple linear regression models was evaluated using **R-square and Root Mean Square Error (RMSE)**.

For **developed countries**, the model produced an **R-square value of 0.1887**, indicating that about **18.87% of the variation in life expectancy** is explained by the selected variables. The **RMSE value of 2.87** suggests that the prediction error is relatively small but the explanatory power of the model is limited.

For **developing countries**, the model showed a much higher **R-square value of 0.7783**, meaning that about **77.83% of the variation in life expectancy** is explained by the independent variables. The **RMSE value of 3.27** indicates a moderate prediction error.

Overall, the results suggest that the **multiple regression model performs better for developing countries**, where socio-economic and health-related factors have a stronger influence on life expectancy.

CONCLUSION

This study aimed to compare life expectancy between developed and developing countries, identify the major determinants influencing it, and predict future life expectancy trends using statistical modeling. Based on data from 2016–2024, significant structural differences were observed between the two groups of countries.

Trend analysis show a statistically significant upward trend in both developed and developing countries. On average, life expectancy increased by approximately 0.39–0.40 years per year in both groups. Over the study period, developed countries experienced a total increase of about 3.03 years, while developing countries showed a slightly higher increase of 3.15 years. However, the interaction model indicates that the rate of improvement does not significantly differ between the two groups, meaning that although both are improving steadily, the life expectancy gap remains persistent.

The One-Way ANOVA results confirmed a statistically significant difference in mean life expectancy between developed and developing countries ($p < 0.001$). Developed countries showed an average life expectancy of approximately 81 years, whereas developing countries averaged around 70 years, indicating a substantial gap of nearly 11 years. The Two-Way ANOVA further demonstrated that both country status and year significantly affect life expectancy, although their interaction was not significant. This indicates that while life expectancy has changed over time, the disparity between developed and developing countries remains persistent.

Correlation analysis revealed that in developing countries, life expectancy is strongly associated with mortality rates, healthcare expenditure, education, and nutritional factors. In contrast, these relationships were comparatively weaker in developed countries, suggesting that developed nations have reached a relatively stable health equilibrium where incremental socio-economic changes produce smaller effects on longevity.

Regression analysis was employed not only to understand the determinants of life expectancy but also to predict future trends. The linear regression model showed moderate explanatory power in developed countries ($R^2 = 0.435$) and very strong explanatory power in developing countries ($R^2 = 0.858$). Similarly, the multiple regression model demonstrated strong predictive capability in developing countries ($R^2 = 0.778$), while the explanatory power

was lower in developed countries ($R^2 = 0.189$), indicating that additional complex factors may influence longevity in advanced economies.

Using these regression model, life expectancy was projected for the next 10 years based on current socio-economic trends. The predictive results suggest that developing countries may experience significant improvements in life expectancy if current improvements in healthcare, education, and economic conditions continue. However, the gap between developed and developing countries is likely to persist unless targeted policy interventions accelerate progress in developing nations.

Overall, this study confirms that life expectancy is strongly linked to socio-economic development and that predictive modeling can provide valuable insights into future trends. The combination of ANOVA, correlation analysis, and regression-based forecasting offers a comprehensive statistical framework for understanding both current disparities and future trajectories of life expectancy.

In conclusion, reducing global inequality in life expectancy requires sustained investment in healthcare infrastructure, reduction of mortality rates, improved education systems, and economic strengthening in developing countries. The predictive results highlight the urgency of policy action to ensure that improvements in longevity are both sustainable and equitable.

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